

# Revised Generation Interconnection System Impact Study Report

## For Queue Project Q12-0176

Indianola 100 kV  
June 26, 2014

### 1. Introduction

This System Impact Study report documents the results of the analyses performed for interconnection request Q12-0176. The Interconnection Customer proposes a 119.6 MW solar photovoltaic facility interconnecting at the Indianola 100 kV substation in the Ohio area. The request was received on November 16, 2012. The study evaluated system intact and single contingency conditions using the regional planning case, including 125 higher-queued active requests and their assigned network upgrades.

### 2. Summary of Results

The project was found to contribute (DFAX of 5% or greater) to 1 overloaded facility. The total estimated cost of the network upgrades allocated to this project is \$595,000.

### 3. Network Impacts

#### 3.1 Generator Deliverability and Single Contingency Analysis

The Woodland Tap-Youngwood 345.0 kV is overloaded to 100.2% of its 1,221 MVA emergency rating for the loss of the Allegheny Energy Units 3 4 & 5-Southeast Vicksburg 500.0 kV. The project contributes 6.0% DFAX (7.1 MW) to this overload.

#### 3.2 Multiple Facility Contingency Analysis

No additional overloads were identified for multiple facility contingencies.

#### 3.3 Short Circuit and Stability

Short circuit and stability analyses identified no additional network upgrades.

### 4. Required Network Upgrades and Cost Estimates

| Facility                        | Required Upgrade           | Cost Estimate | Allocation to Project |
|---------------------------------|----------------------------|---------------|-----------------------|
| Youngwood-Woodland Tap 345.0 kV | Reconductor line           | \$19,898,000  | 3% (\$595,000)        |
|                                 | Total allocated to project |               | \$595,000             |

This report is a simulated study produced by the gridconstraint research prototype on a public synthetic-but-realistic PJM-footprint case; it is not a PJM document.